14. Strings

In java, String is class and a non-primitive data type which allows alpha numeric, words and statements. String can be created in java using two methods →

1. ***Using literals →*** *This is exactly same as syntax used for creating the variable from primitive datatype.*

```
String name = "Java";
```

2. ***Using new keyword*** → Same as syntax used for creating object. String class has a parameterized constructor with parameter as a String.

```
String name = new String("Java");
```

Difference between String created using literals and using new keyword.

In JVM, there is a memory called as heap memory which stores objects. In heap memory there is a part called as String constant pool (SCP).

When Strings are created using literals →

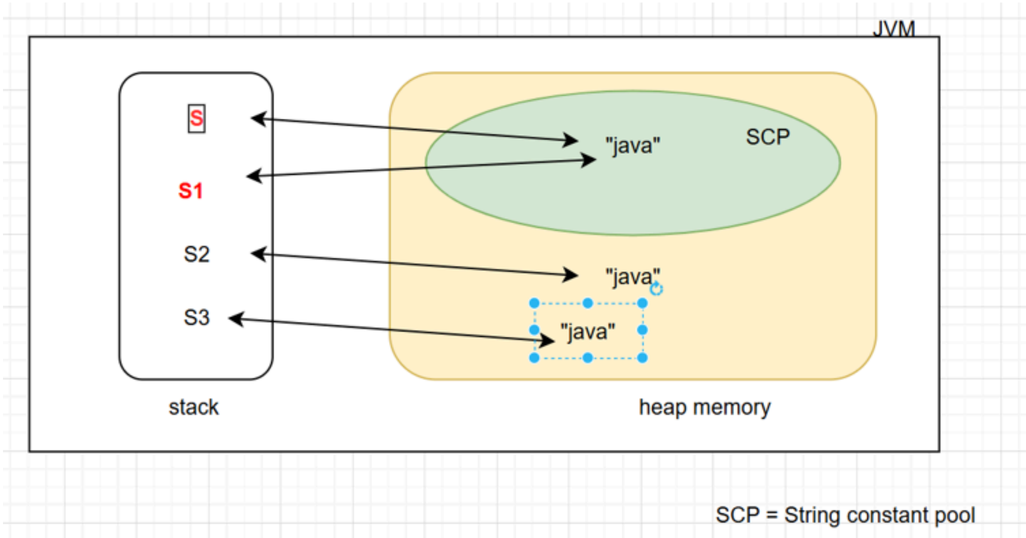
- When Strings are created using literals, variables are stored in String constant pool which is subset of heap memory.
- Advantage of String constant pool is if two Strings with same value(consider variable name s and s1) are created then only one memory is allocated to it and both s and s1 objects (s and s1 are stored in stack area) will point to same memory.(Detailed in diagram).

When Strings are created using new keyword →

- When Strings are created using a new keyword, a new object is created and is stored in heap memory but not in string constant pool.
- Resulting into new memory allocation every time a new keyword is used for creating a String. Though Strings have same value in it, different memory will be allocated for different reference variable.(Detailed in diagram).

Following diagram shows difference -

- Here S and Sa1 are created using literals and are stored in SCP.
- S2 and S3 are created using *new* keyword and stored in heap memory.



Way to create String	Heap area	SCP
using *literals*	❌	✅
using *new* keyword	✅	❌

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